Assam flood 2016 (radar, M8)

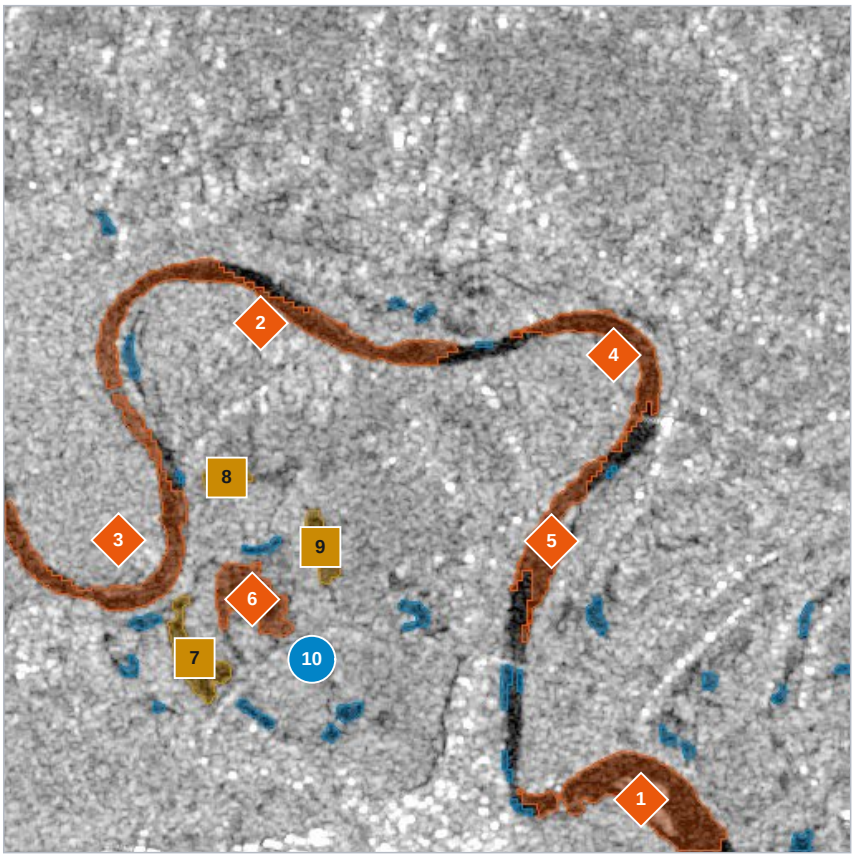
Brahmaputra floodplain, Assam, India | Optical - Sentinel-2 - 2016-08-12 (same day) to Radar - Sentinel-1 VV - 2016-08-12

What we found

We found 37 real flooded areas covering 139 hectares. The biggest and strongest (rank 1) is new water / flooding: 24.42 ha at 26.5873, 93.8159.

By priority: 0 Critical, 6 High, 3 Medium, 28 Low.

The detector first flagged 3,490 candidate areas. The rest were removed as false alarms: permanent rivers and lakes, radar speckle, areas below the minimum size.



Numbers = rank of the 10 highest-priority changes. Outline colour = priority.

Changes by priority

Critical	0
High	6
Medium	3
Low	28

Totals

Area changed: 139 ha

How to read this report

Rank	Every change has a number. Rank 1 is the highest priority: the largest and strongest change. Priority = size x strength of the change.
Priority colour and shape	Critical (octagon): 50 ha or more of flood water; High (diamond): 10 ha or more; Medium (square): 2 ha or more; Low (circle): smaller than 2 ha.
Confidence	How sure the system is that the change is real, from 0 to 100 %. Changes marked 'needs a human check' sit between 50 % and 70 %.
What changed	New construction, vegetation cleared, vegetation growth, new water / flooding, water receded, or unexplained change. These come from rules on the satellite colours, not from knowing what is actually on the ground.
Raw detections	Everything the detector first flagged. Most are not real (clouds, crop seasons, shadows, specks), so only changes that pass the filters are listed here.

The 10 highest-priority changes

Start here. Each card says where the change is, how big it is, why it was flagged and what to do.

1

New water / flooding

Rank 1 of 37 | High priority

Size	24.42 ha (244,240 m2), about 34 football pitches
Confidence	99 %
Location	26.58725, 93.81585 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -19.9 dB, 5.9 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check soon. A substantial change that is probably real.

Before



After



2

New water / flooding

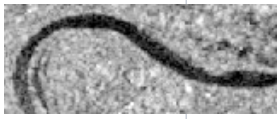
Rank 2 of 37 | High priority

Size	23.37 ha (233,683 m2), about 33 football pitches
Confidence	99 %
Location	26.61314, 93.79520 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -19.0 dB, 4.9 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check soon. A substantial change that is probably real.

Before



After



3

New water / flooding

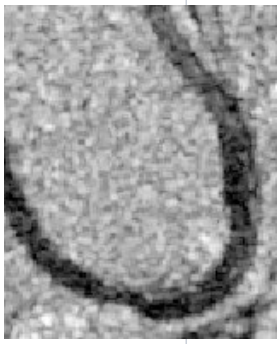
Rank 3 of 37 | High priority

Size	21.15 ha (211,457 m2), about 30 football pitches
Confidence	99 %
Location	26.60135, 93.78747 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -18.1 dB, 4.0 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check soon. A substantial change that is probably real.

Before



After



4

New water / flooding

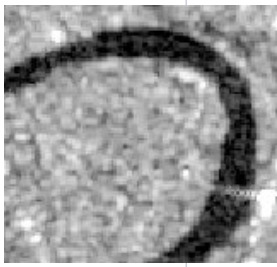
Rank 4 of 37 | High priority

Size	12.48 ha (124,764 m2), about 17 football pitches
Confidence	99 %
Location	26.61140, 93.81436 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -19.8 dB, 5.7 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check soon. A substantial change that is probably real.

Before



After



5

New water / flooding

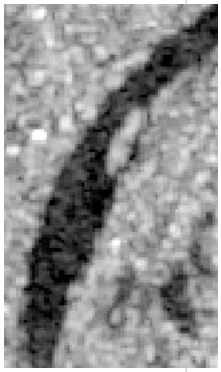
Rank 5 of 37 | High priority

Size	11.63 ha (116,320 m2), about 16 football pitches
Confidence	99 %
Location	26.60130, 93.81099 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -19.3 dB, 5.2 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check soon. A substantial change that is probably real.

Before



After



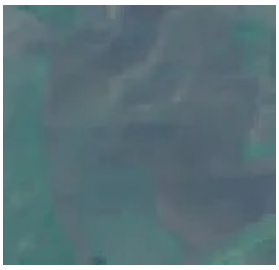
6

New water / flooding

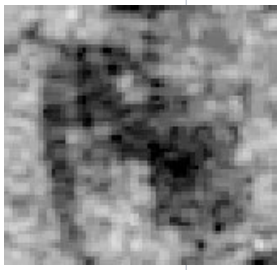
Rank 6 of 37 | High priority

Size	10.73 ha (107,334 m2), about 15 football pitches
Confidence	93 %
Location	26.59814, 93.79474 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -17.5 dB, 3.4 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check soon. A substantial change that is probably real.

Before



After



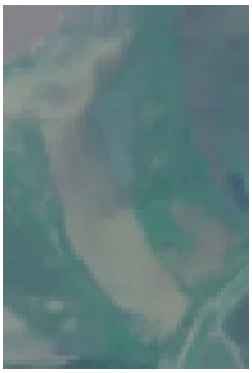
7

New water / flooding

Rank 7 of 37 | Medium priority

Size	6.97 ha (69,689 m2), about 10 football pitches
Confidence	89 %
Location	26.59495, 93.79161 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -17.1 dB, 3.1 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check when you can, for example on a routine inspection.

Before



After



8

New water / flooding

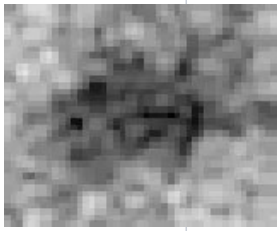
Rank 8 of 37 | Medium priority

Size	3.61 ha (36,132 m2), about 5 football pitches
Confidence	82 %
Location	26.60475, 93.79335 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -16.6 dB, 2.6 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check when you can, for example on a routine inspection.

Before



After



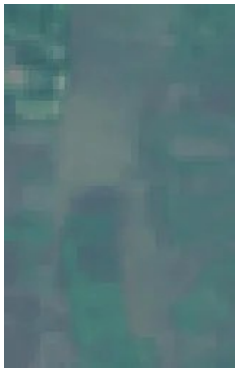
9

New water / flooding

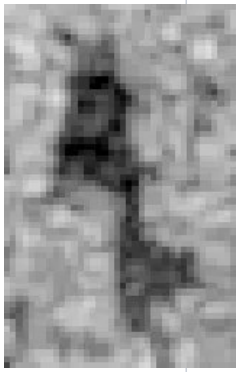
Rank 9 of 37 | Medium priority

Size	3.44 ha (34,353 m2), about 5 football pitches
Confidence	96 %
Location	26.60098, 93.79844 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -17.7 dB, 3.7 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Check when you can, for example on a routine inspection.

Before



After



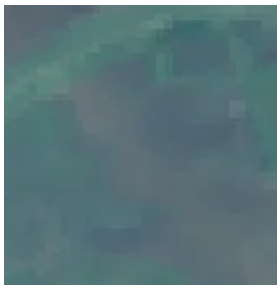
10

New water / flooding

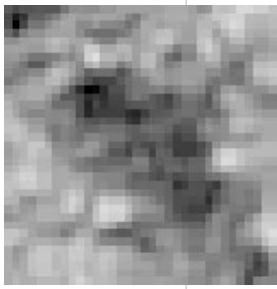
Rank 10 of 37 | Low priority

Size	1.74 ha (17,355 m2), about 2 football pitches
Confidence	73 %
Location	26.59488, 93.79797 (open in Google Maps)
What it is	Ground that was dry is now covered by water.
Why flagged	radar backscatter -15.9 dB, 1.8 dB below the event water threshold (-14.1 dB); grown from a seed 3 dB darker; not permanent water (JRC)
What to do	Small or weak change. Often noise: confirm before acting.

Before



After



All 37 changes found

Every change that survived the false-alarm filters, in priority order (rank 1 is the highest). A * after the confidence means it needs a human check. The CSV and GeoPackage downloads hold the same list with every attribute.

Rank	Priority	What changed	Size (ha)	Confidence	Latitude	Longitude
1	High	New water / flooding	24.42	99 %	26.58725	93.81585
2	High	New water / flooding	23.37	99 %	26.61314	93.79520
3	High	New water / flooding	21.15	99 %	26.60135	93.78747
4	High	New water / flooding	12.48	99 %	26.61140	93.81436
5	High	New water / flooding	11.63	99 %	26.60130	93.81099
6	High	New water / flooding	10.73	93 %	26.59814	93.79474
7	Medium	New water / flooding	6.97	89 %	26.59495	93.79161
8	Medium	New water / flooding	3.61	82 %	26.60475	93.79335
9	Medium	New water / flooding	3.44	96 %	26.60098	93.79844
10	Low	New water / flooding	1.74	73 %	26.59488	93.79797
11	Low	New water / flooding	1.58	82 %	26.59727	93.80364
12	Low	New water / flooding	1.55	87 %	26.59718	93.81350
13	Low	New water / flooding	1.17	99 %	26.59200	93.80008
14	Low	New water / flooding	1.14	89 %	26.59187	93.79499
15	Low	New water / flooding	1.14	70 %	26.61118	93.78816
16	Low	New water / flooding	1.13	98 %	26.59344	93.80856
17	Low	New water / flooding	1.09	99 %	26.58493	93.82463
18	Low	New water / flooding	1.01	79 %	26.59689	93.78892
19	Low	New water / flooding	0.97	79 %	26.60091	93.79538
20	Low	New water / flooding	0.76	66 % *	26.59697	93.82480
21	Low	New water / flooding	0.75	69 % *	26.59439	93.78812
22	Low	New water / flooding	0.69	92 %	26.61359	93.80414
23	Low	New water / flooding	0.65	72 %	26.59064	93.81731
24	Low	New water / flooding	0.65	71 %	26.61845	93.78681
25	Low	New water / flooding	0.63	68 % *	26.59364	93.81958
26	Low	New water / flooding	0.58	94 %	26.59082	93.79908
27	Low	New water / flooding	0.52	95 %	26.58677	93.80936
28	Low	New water / flooding	0.52	99 %	26.58903	93.80861
29	Low	New water / flooding	0.49	80 %	26.59291	93.82342
30	Low	New water / flooding	0.47	74 %	26.58993	93.81844
31	Low	New water / flooding	0.36	78 %	26.61415	93.80265
32	Low	New water / flooding	0.35	97 %	26.59368	93.80931
33	Low	New water / flooding	0.31	75 %	26.59226	93.78961
34	Low	New water / flooding	0.30	71 %	26.59431	93.82681
35	Low	New water / flooding	0.27	99 %	26.61192	93.80738
36	Low	New water / flooding	0.21	96 %	26.60499	93.81429
37	Low	New water / flooding	0.21	78 %	26.60475	93.79087

Data sources

- Copernicus Sentinel-1 IW GRD (dB) and Sentinel-2, via Sen1Floods11 v1.1 (CC-BY 4.0)
- JRC Global Surface Water (permanent water mask)

Accuracy check

On this scene the flood extent was compared with expert hand-drawn labels (Sen1Floods11 expert hand labels (flood water, JRC permanent water excluded)). Overlap (IoU): 0.43 for this method, 0.42 for a simple fixed threshold. This is one demonstration scene, not a held-out test.

Limits - what this report cannot tell you

- Open-water flood extent estimate: flooded crops and streets can look bright on radar and may be missed.
- Single date shown; the full M8 compares before/after radar scenes.
- Impact layers are not computed for this 2016 test chip
- This report points to where to look. Confirm important changes on the ground before acting.